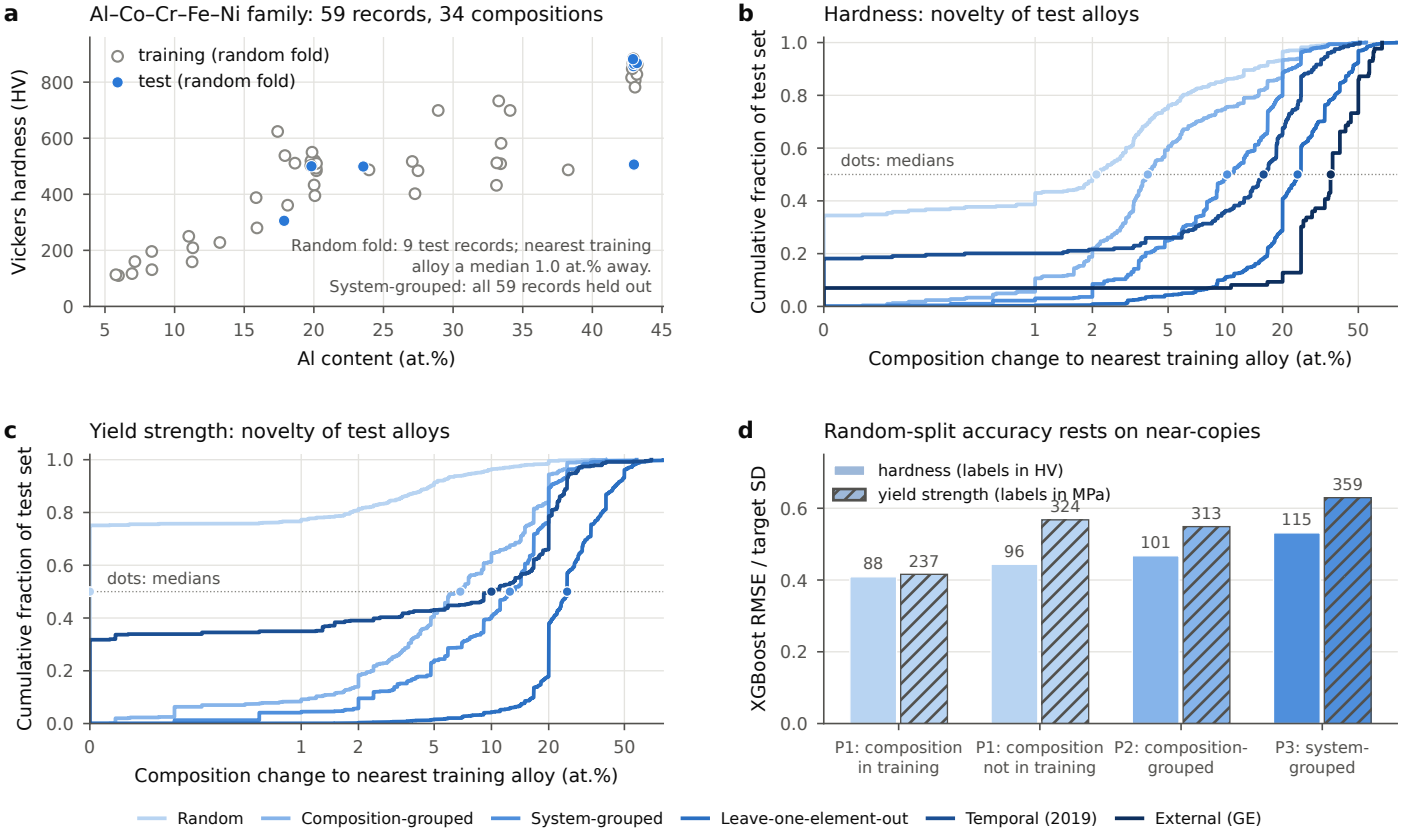
a
Al–Co–Cr–Fe–Ni family: 59 records, 34 compositions
training (random fold)
test (random fold)
Random fold: 9 test records; nearest training alloy a median 1.0 at.% away.
System-grouped: all 59 records held out
Vickers hardness (HV)
Al content (at.%)
b
Hardness: novelty of test alloys
dots: medians
Cumulative fraction of test set
Composition change to nearest training alloy (at.%)
c
Yield strength: novelty of test alloys
dots: medians
Cumulative fraction of test set
Composition change to nearest training alloy (at.%)
d
Random-split accuracy rests on near-copies
hardness (labels in HV)
yield strength (labels in MPa)
XGBoost RMSE / target SD
88
237
96
324
101
313
115
359
P1: composition in training
P1: composition not in training
P2: composition-grouped
P3: system-grouped
Random
Composition-grouped
System-grouped
Leave-one-element-out
Temporal (2019)
External (GE)